SNP-Based Analysis using *plink* and R

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Single nucleotide polymorphisms (SNPs)

What is a SNP?

- A SNP is a position in the genome where a single nucleotide base differs between individuals.

~4–5 million SNPs in a typical human genome

~1 per 1,000 bp average spacing along the genome

Common = MAF > 1–5% the variant is frequent in the population

Two people, one position

PERSON A

A C G T **A** G C T

PERSON B

A C G T **G** G C T

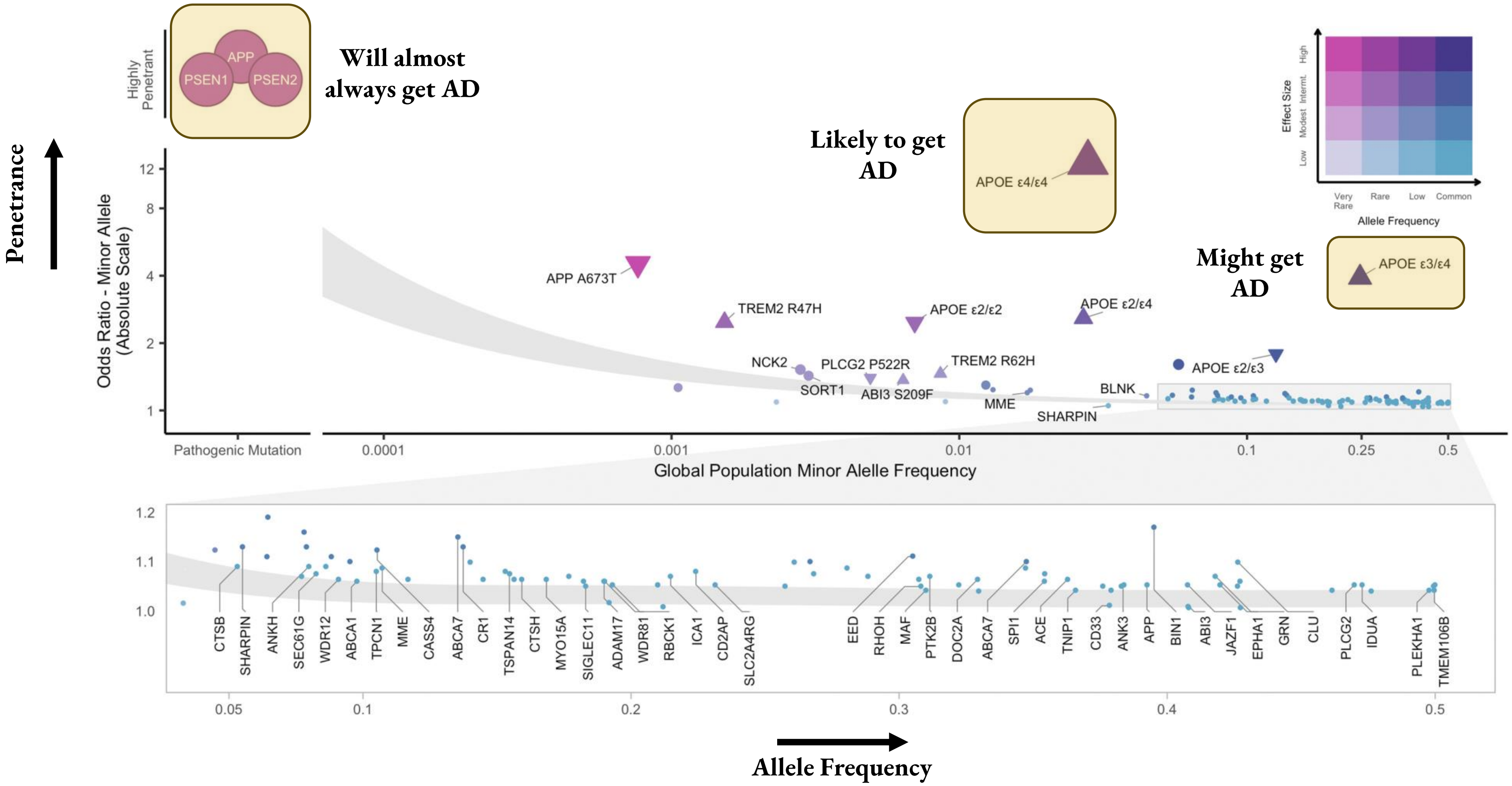
This single-base difference (A / G) is the SNP. The two versions are called alleles.

Types of SNPs

- The location of the SNP doesn't really matter, but the type of SNP can have big implications. Types of SNPs include:
  - Coding
    - Missense: can alter protein function
    - Synonymous: often silent
    - Nonsense: truncates a protein
  - Non-Coding
    - Regulatory: tunes gene expression
    - Splice-site: disrupts transcript assembly
  - Intergenic
    - This is the most common SNP type, causes subtle effects and trans regulation

SNPs in Alzheimer's disease

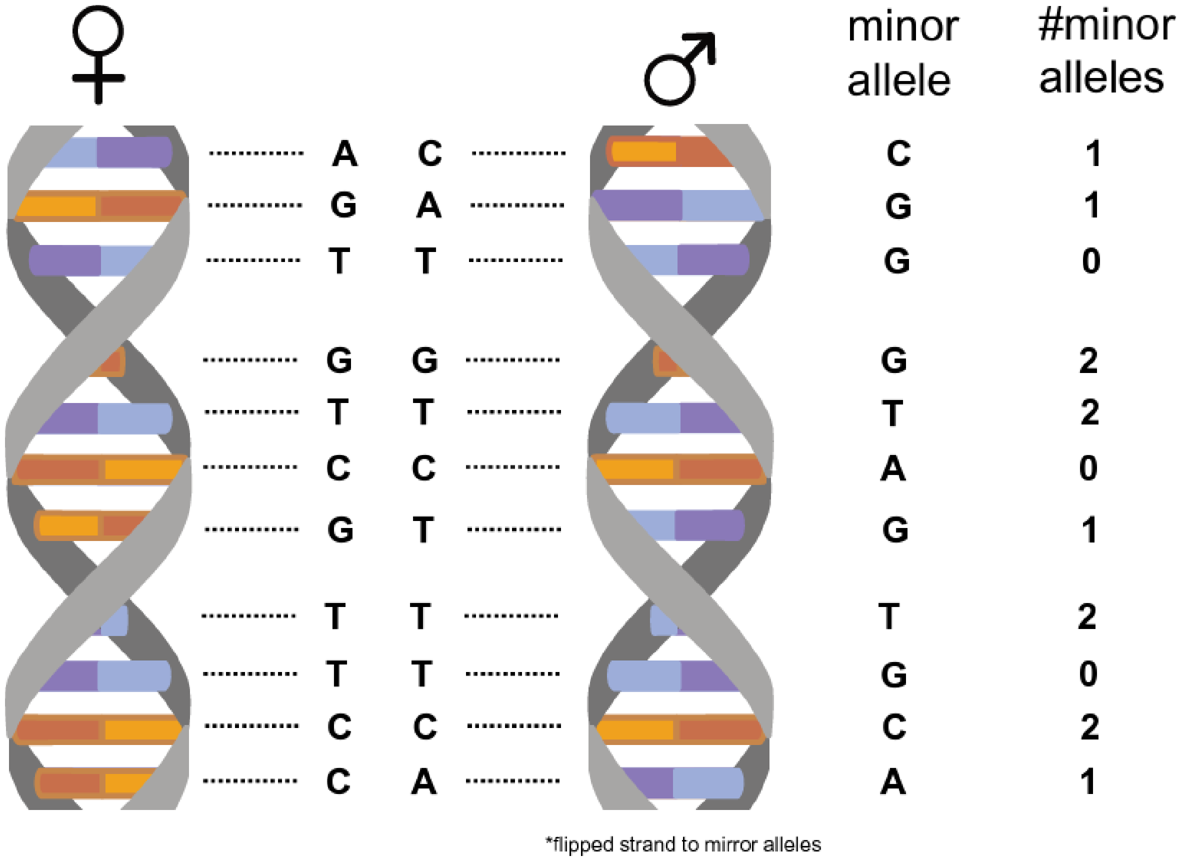
- 100+ SNPs have been identified in AD



Occurrence of multiple risk genes increases odd of AD

The Number of SNPs Can Matter

- The dosage of an allele mirrors the number of copies from both inherited DNA.
  - It therefore ranges from 0-2.



SNP-Based Analysis

- SNP-based analysis is usually logistic or linear regression analysis
  - The SNP can be coded as “carrier/non-carrier” or “additive”.
- Logistic regression example:
  - $AD\ Diagnosis \sim SNP + Age + Sex + PC1 + PC2 + PC3$
- Linear regression example:
  - $Hippocampal\ volume \sim SNP + Age + Sex + PC1 + PC2 + PC3$
- Some common SNPs in AD research

SNP	Chromosome	BP	Minor Allele Frequency	Previously Associated Gene
rs429358	19	44908684	0.074	*APOE*
rs6733839	2	127135234	0.390	*BIN1*
rs4844610	1	207629207	0.171	*CR1*

SNP-Based Analysis Walkthrough

Step 1: Extract SNP dosages from plink dataset

- In command line, type:

```
### Set some variables
PLINK=./plink/plink
BFILE=data/files-imputation/postImputation/ADNI1.impQC.rs
OUTDIR=my-analysis

### Navigate to home directory
cd ~

### Create directory and generate dosage file for APOE SNP
mkdir $OUTDIR
$PLINK --bfile $BFILE --recode A --snps rs429358,rs769449 --out $OUTDIR/ADNI1_APOE_snps
```

- This will generate a .raw file of dosages for each participants.
- NOTE: You can also input a large list of SNPs as a text file.

```
(base) [as2-streaming-user@ip-198-19-4-5 my-analysis]$ head ADNI1_APOE_snps.raw
FID IID PAT MAT SEX PHENOTYPE rs769449_A rs429358_C
1 014_S_0520 0 0 0 -9 2 2
2 005_S_1341 0 0 0 -9 1 1
3 012_S_1175 0 0 0 -9 0 0
4 012_S_0803 0 0 0 -9 0 0
5 018_S_0055 0 0 0 -9 0 0
6 027_S_0118 0 0 0 -9 0 0
7 027_S_0403 0 0 0 -9 0 0
8 053_S_0389 0 0 0 -9 0 1
9 041_S_0262 0 0 0 -9 0 0
```

Step 2: Read in data

- In R, type:

```
### Load libraries
library(tidyverse)
library(here)
library(broom)

### Create variables for files and directories
genofile <- here("~","my-analysis","ADNI1_APOE_snps.raw")
phenofile <- here("~","data","ADNI1_demographics_immersive.txt")
outdir <- here("~","my-analysis")

### Import and preview genotype data
geno <- read_table(genofile)
glimpse(geno)

### Import and preview phenotype data
pheno <- read_table(phenofile)
glimpse(pheno)
```

```
> glimpse(geno)
Rows: 757
Columns: 8
$ FID       <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, …
$ IID       <chr> "014_S_0520", "005_S_1341", "012_S_1175", "012_S_0803", "01…
$ PAT       <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,…
$ MAT       <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,…
$ SEX       <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,…
$ PHENOTYPE <dbl> -9, -9, -9, -9, -9, -9, -9, -9, -9, -9, -9, -9, -9, -9, -9,…
$ rs769449_A <dbl> 2, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 1, 0, 0, 1, 1, 0, 0,…
$ rs429358_C <dbl> 2, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 2, 0, 1, 0, 0, 1, 1, 1, 0,…
```

```
> glimpse(pheno)
Rows: 757
Columns: 19
$ ID2                 <chr> "0002", "0003", "0004", "0005", "0006", "0007", "0…
$ ID                  <chr> "011_S_0002", "011_S_0003", "022_S_0004", "011_S_0…
$ STUDY               <chr> "ADNI1", "ADNI1", "ADNI1", "ADNI1", "ADNI1", "ADNI…
$ CODE                <chr> "bl", "bl", "bl", "bl", "bl", "bl", "bl", "bl", "b…
$ EXAMDATE            <chr> "08/09/2005", "12/09/2005", "08/11/2005", "07/09/2…
$ DISEASE_STATUS      <chr> "CN", "AD", "LMCI", "CN", "LMCI", "AD", "CN", "AD"…
$ AGE                 <dbl> 74.3, 81.3, 67.5, 73.7, 80.4, 75.4, 84.5, 73.9, 78…
$ REPORTED_SEX        <chr> "Male", "Male", "Male", "Male", "Female", "Male", …
$ YEARS_EDUCATION     <dbl> 16, 18, 10, 16, 13, 10, 18, 12, 12, 18, 9, 18, 18,…
$ APOE4               <dbl> 0, 1, 0, 0, 0, 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 0,…
$ C1                  <dbl> -0.00964356, -0.01185320, 0.01839370, -0.00944065,…
$ C2                  <dbl> -0.004582260, -0.004940880, -0.000411387, -0.00248…
$ C3                  <dbl> 0.00624303, 0.00976110, -0.00737629, 0.00653403, 0…
$ C4                  <dbl> -0.021170900, 0.000194991, 0.005400770, 0.00083977…
$ ATgroup             <chr> NA, "A+T-", NA, "A+T+", NA, NA, "A-T+", "A+T+", "A…
$ tesla               <dbl> NA, 1.5, NA, 1.5, 1.5, 1.5, 1.5, 1.5, 1.5, 1.5, 1.…
$ ICV                 <dbl> NA, 1951081, NA, 1704436, 1483142, 1370389, 142442…
$ Whole_hippocampus   <dbl> NA, 5649.286, NA, 6206.037, 4689.231, 5439.904, 53…
$ DISEASE_STATUS_DICO <dbl> 0, 1, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0,…
```


Walkthrough courtesy of Yuetiva Robles

Step 2: Subset to columns of interest

- In R, type:

```
### Extract IDs and SNP data
geno_sub <- geno %>%
  select(FID, IID, rs769449_A, rs429358_C)

### Combine genotype data, age, sex, AD status, HV, ICV, and PCs for analyses
hv <- pheno %>%
  select("IID"  = ID, "Diagnosis" = DISEASE_STATUS,
         "ADstatus" = DISEASE_STATUS_DICO,
         AGE, "SEX" = REPORTED_SEX, "HV" = Whole_hippocampus,
         ICV, C1, C2, C3, C4) %>%
  right_join(geno_sub, by = "IID")
glimpse(hv)
```

```
> glimpse(hv)
Rows: 757
Columns: 14
$ IID        <chr> "011_S_0002", "011_S_0003", "022_S_0004", "011_S_0005", "10…
$ Diagnosis  <chr> "CN", "AD", "LMCI", "CN", "LMCI", "AD", "CN", "AD", "CN", "…
$ ADstatus   <dbl> 0, 1, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0,…
$ AGE        <dbl> 74.3, 81.3, 67.5, 73.7, 80.4, 75.4, 84.5, 73.9, 78.5, 80.8,…
$ SEX        <chr> "Male", "Male", "Male", "Male", "Female", "Male", "Female",…
$ HV         <dbl> NA, 5649.286, NA, 6206.037, 4689.231, 5439.904, 5369.138, 4…
$ ICV        <dbl> NA, 1951081, NA, 1704436, 1483142, 1370389, 1424427, 149913…
$ C1         <dbl> -0.00964356, -0.01185320, 0.01839370, -0.00944065, -0.01147…
$ C2         <dbl> -0.004582260, -0.004940880, -0.000411387, -0.002487880, -0.…
$ C3         <dbl> 0.00624303, 0.00976110, -0.00737629, 0.00653403, 0.00474530…
$ C4         <dbl> -0.021170900, 0.000194991, 0.005400770, 0.000839774, 0.0034…
$ FID        <dbl> 663, 669, 535, 80, 138, 642, 67, 403, 633, 332, 190, 130, 3…
$ rs769449_A <dbl> 0, 1, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0,…
$ rs429358_C <dbl> 0, 1, 0, 0, 0, 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 0,…
```

Step 2: Run regression analysis in R

- Now we can run some models. Let's do analysis on hippocampal volume:

```
### Run Main Model (rs429358)
rs429358hv.mod <- lm(
  HV ~ rs429358_C + AGE + SEX + ICV,
  data = hv)
summary(rs429358hv.mod)
tidy(rs429358hv.mod)

### Run Model with LD SNP (rs769449)
rs769449hv.mod <- lm(
  HV ~ rs769449_A + AGE + SEX + ICV,
  data = hv)
summary(rs769449hv.mod)
tidy(rs769449hv.mod)
```

Term	Estimate	Std. Error	Statistic	P-value
(Intercept)	6299	630	10.0	9.11×10^{-22}
rs429358_C	-480	58.4	-8.23	1.37×10^{-15}
AGE	-40.2	6.00	-6.71	4.80×10^{-11}
SEXMale	277	96.1	2.88	4.17×10^{-3}
ICV	0.0015	0.0003	5.38	1.09×10^{-7}

Term	Estimate	Std. Error	Statistic	P-value
(Intercept)	5791	632	9.17	9.41×10^{-19}
rs769449_A	-423	61.7	-6.86	1.84×10^{-11}
AGE	-36.0	6.03	-5.97	4.16×10^{-9}
SEXMale	257	97.6	2.63	8.79×10^{-3}
ICV	0.0016	0.0003	5.54	4.55×10^{-8}

Step 2: Run regression analysis in R

- Now let's look at AD case/control status:

```
### Run Main Model (rs429358)
rs429358ad.mod <- glm(
  ADstatus ~ rs429358_C + AGE + SEX,
  data = hv,   family = binomial)
summary(rs429358ad.mod)
tidy(rs429358ad.mod)

### Run Model with LD SNP (rs769449)
rs769449ad.mod <- glm(
  ADstatus ~ rs769449_A + AGE + SEX,
  data = hv,   family = binomial)
summary(rs769449ad.mod)
tidy(rs769449ad.mod)
```

Term	Estimate	Std. Error	Statistic	P-value
(Intercept)	-3.35	1.06	-3.14	0.00167
rs429358_C	0.794	0.133	5.95	0.00000000263
AGE	0.0246	0.0138	1.78	0.0751
SEXMale	-0.393	0.180	-2.19	0.0288

Term	Estimate	Std. Error	Statistic	P-value
(Intercept)	-2.72	1.04	-2.62	0.00882
rs769449_A	0.664	0.136	4.90	0.000000983
AGE	0.0185	0.0136	1.36	0.173
SEXMale	-0.373	0.178	-2.10	0.0358

Now you're ready to expand to the whole genome!